Patient Condition–Aware Drug Recommendation and Clinical Review Sentiment Analysis Using NLP and Machine Learning

Improving Healthcare Decisions Using NLP and Machine Learning

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ABSTRACT

This study proposes a patient condition–aware drug recommendation system that integrates Machine Learning and Natural Language Processing (NLP). The system analyzes structured patient data such as symptoms, diagnosis, age, and medical history to suggest suitable medications. In addition, patient drug reviews are analyzed using NLP techniques to determine sentiment as positive, negative, or neutral.Text preprocessing methods like tokenization, stopword removal, and TF-IDF vectorization are applied before training the sentiment classification models. By combining clinical prediction results with patient review sentiment, the system provides more accurate and trustworthy drug recommendations. This approach supports better treatment decisions and helps in developing more personalized and patient-centered healthcare solutions.

INTRODUCTION

The healthcare field produces a large amount of data such as patient records, treatment history, and drug reviews. Choosing the right medication for a patient is difficult because drug effectiveness can vary based on factors like symptoms, age, and medical history. This project proposes a Patient Condition–Aware Drug Recommendation System that uses Machine Learning and Natural Language Processing to analyze patient conditions and clinical reviews to suggest suitable medications.

By combining patient condition analysis with sentiment analysis of drug reviews, the system provides more accurate and reliable recommendations. This approach helps improve healthcare decision support and supports more personalized treatment for patients.

EXISTING SYSTEM

- Existing drug recommendation systems rely mainly on **clinical guidelines** and **structured patient data** such as symptoms, diagnosis, and medical history.
- These systems mostly use **rule-based approaches** to suggest medications.
- **Patient reviews and feedback** are usually ignored in the recommendation process.
- Traditional systems **do not analyze unstructured text data** like clinical reviews.
- As a result, these systems provide **general recommendations** with limited personalization.

PROBLEM STATEMENT

- Selecting the **most suitable drug** is difficult due to many available medications and varying patient conditions.
- Existing systems rely mainly on **clinical guidelines and structured patient data**, limiting personalized treatment.
- **Patient reviews and clinical feedback** containing useful drug information are often ignored.
- Traditional systems **cannot analyze unstructured text** or patient sentiments.
- Therefore, there is a need for an **ML and NLP-based patient condition–aware system** to provide accurate and personalized drug recommendations.

SYSTEM REQUIREMENTS

SOFTWARE REQUIREMENTS

- **Programming:** Python 3.x, Pandas, NumPy, Scikit-learn
- **NLP & Text Processing:** NLTK, spaCy,TF-IDF Vectorizer
- **Storage:** MySQL, SQLite, CSV, Excel
- **Deployment:** Flask, Django

HARDWARE REQUIREMENTS

- System with minimum 4–8 GB RAM
- Multi-core processor (Intel i5 or above recommended)
- Storage: Minimum 10 GB free space
- GPU (optional but recommended for deep learning models)
- Internet connection for accessing datasets and deployment

PROPOSED SYSTEM

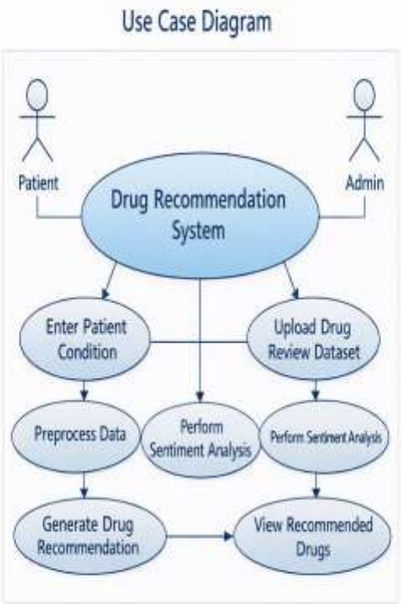
- Uses Machine Learning and NLP for intelligent drug recommendation
- Analyzes both structured (patient data) and unstructured data (reviews)
- Provides personalized drug suggestions based on patient condition
- Incorporates sentiment analysis to improve recommendation accuracy
- Highlights drug interactions and possible side effects
- Continuously learns from new data and user feedback
- Improves decision-making and treatment outcomes

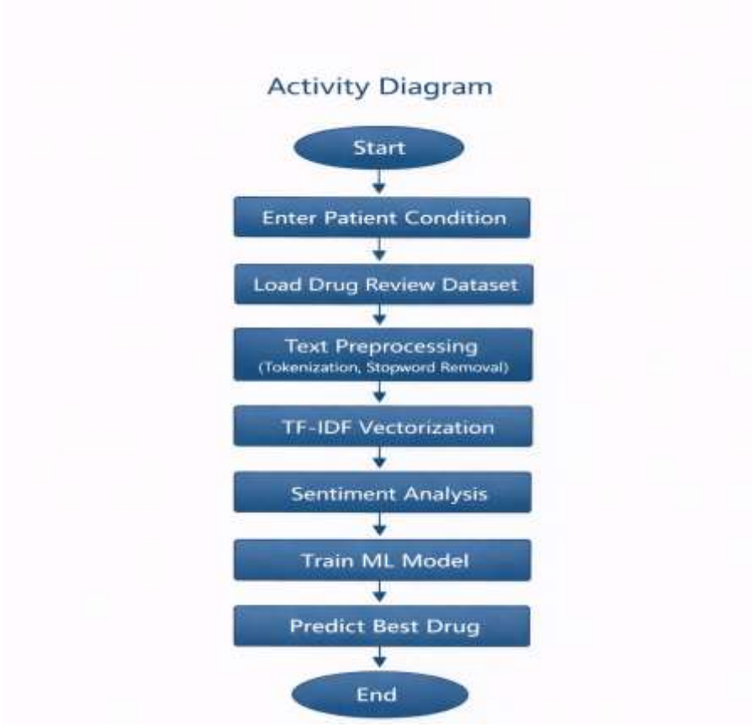
MODEL & ALGORITHMS

- **Classification Models:** Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM)
- **Drug Ranking Algorithms:** Weighted Scoring, Collaborative Filtering
- **NLP Techniques:** Tokenization, Stopword Removal, Lemmatization, TF-IDF Vectorization
- **Sentiment Analysis Models:** Logistic Regression, Naive Bayes, LSTM, CNN
- **Integration:** Combine patient condition predictions with sentiment scores to generate the final ranked drug recommendations

UML DIAGRAMS

A **UML (Unified Modeling Language)** diagram is a standardized visual representation of a software system's architecture, design, and implementation

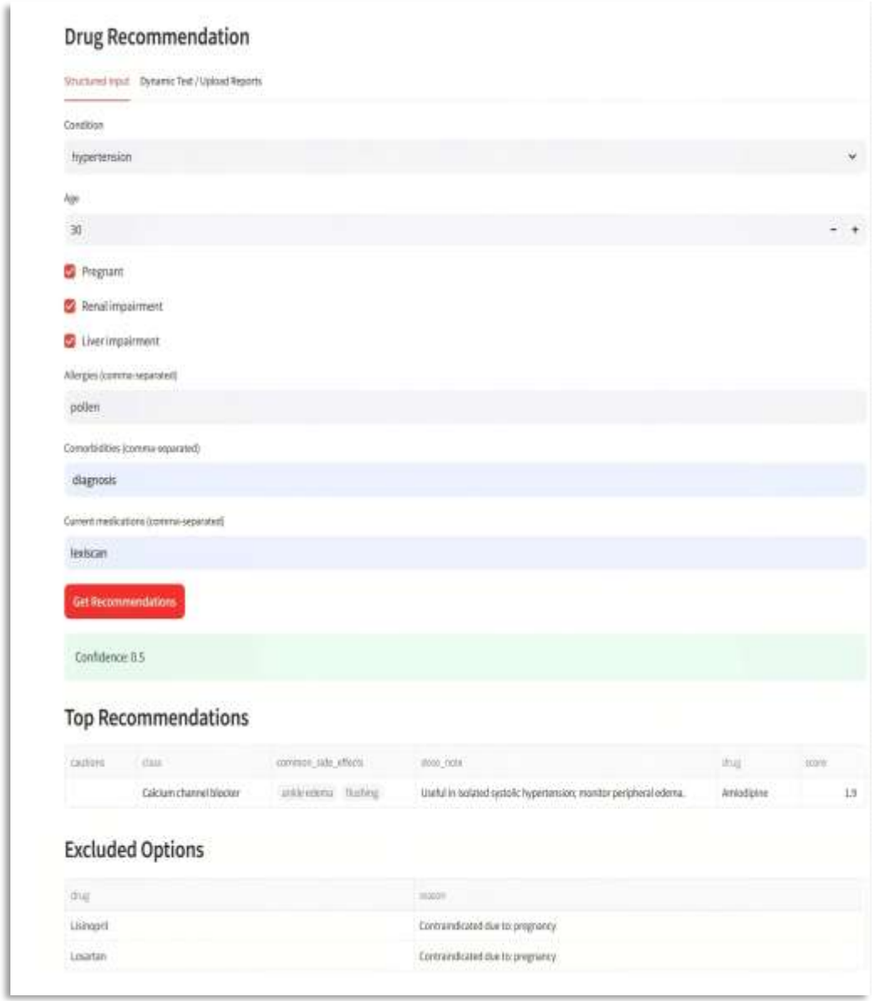


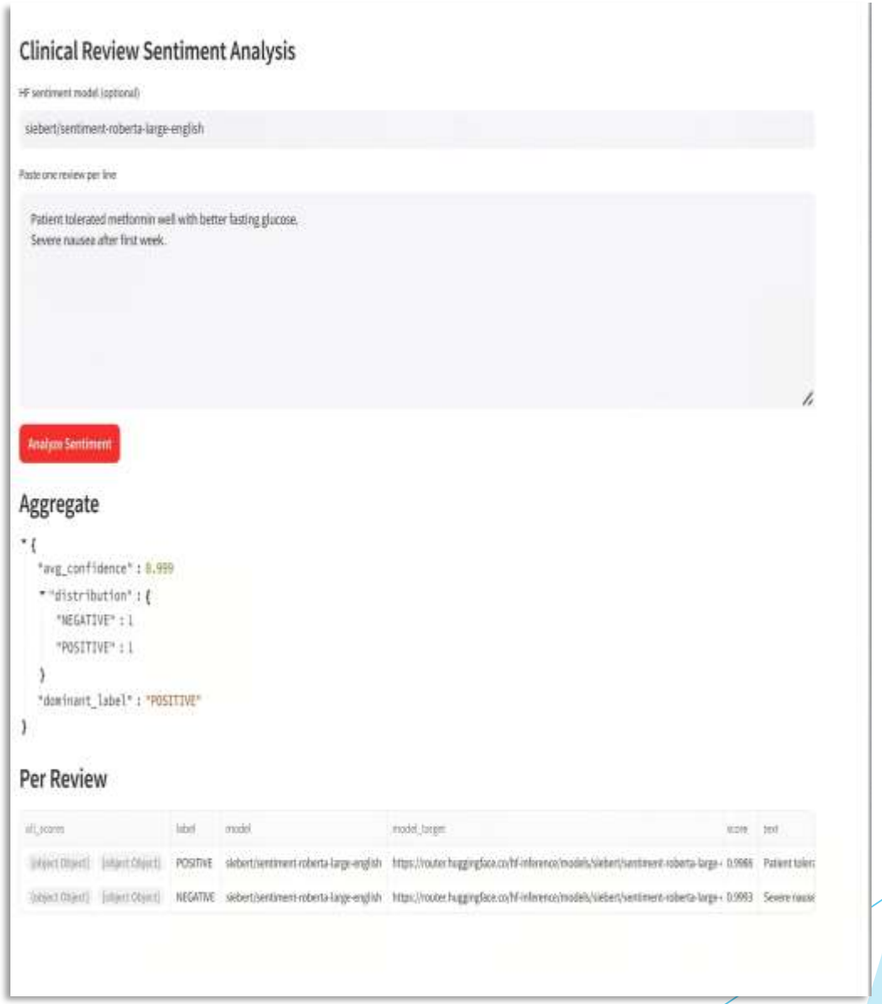


Shows how the patient interacts with the drug recommendation system to input their condition and receive drug suggestions.

Shows the step-by-step workflow of the system from entering patient data to generating the recommended drug.

OUTPUT


Drug Recommendation
Structured Input
Dynamic Text / Upload Reports
Condition
hypertension
Age
30
Pregnant
Renal impairment
Liver impairment
Allergies (comma-separated)
pollen
Comorbidities (comma-separated)
diagnosis
Current medications (comma-separated)
lexiscan
Get Recommendations
Confidence: 0.5
Top Recommendations
Calcium channel blocker
Useful in isolated systolic hypertension; monitor peripheral edema.
Amlodipine
1.9
Excluded Options
Lisinopril
Contraindicated due to pregnancy
Losartan
Contraindicated due to pregnancy


Clinical Review Sentiment Analysis
HF sentiment model (optional)
siebert/sentiment-roberta-large-english
Paste one review per line
Patient tolerated metformin well with better fasting glucose.
Severe nausea after first week.
Analyze Sentiment
Aggregate
{
"avg_confidence": 0.999
"distribution": {
"NEGATIVE": 1
"POSITIVE": 1
}
"dominant_label": "POSITIVE"
}
Per Review
POSITIVE
siebert/sentiment-roberta-large-english
NEGATIVE
siebert/sentiment-roberta-large-english

TESTING

- The system was tested using **unit, integration, system, performance, and validation testing** to ensure accuracy and reliability.
- **Unit testing** verified data preprocessing, model prediction, and sentiment classification.
- **Integration testing** ensured smooth interaction between recommendation and sentiment modules.
- **System testing** validated the complete workflow.
- **Performance testing** evaluated training time, prediction speed, and scalability.
- **Cross-validation** confirmed consistent accuracy, precision, recall, and F1-score.

CONCLUSION

This study presents a patient condition–aware drug recommendation system that combines machine learning with sentiment analysis to improve medication selection. By analyzing both patient medical data and drug reviews, the system provides personalized and data-driven recommendations. It processes large amounts of feedback and continuously improves its predictions, helping healthcare professionals and patients make better treatment decisions.

FUTURE SCOPE

- Integrate with **real-time hospital records** for instant drug recommendations.
- Support **multiple health conditions** using advanced deep learning models.
- Include **lifestyle, genetic, and demographic data** for better personalization.
- Develop **mobile and cloud-based platforms** for easy access.
- Add **multi-language support** and integration with **wearable devices** for real-time monitoring.
- Provide **explainable recommendations** and generate reports to assist doctors.

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